

A LONGITUDINAL HQIV/O-MAXWELL CHANNEL FOR CORONAL HEATING: BOUNDARY-FORM STRESS ON PHOTOSPHERE-TO-CORONA FLUX TUBES

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ABSTRACT. Background. A companion note [?] gives an HQIV/O-Maxwell account of the longitudinal electromagnetic stress reported by recent direct-measurement experiments on current-carrying metallic wires [?]. The proposed residual force is an axial component of the inhomogeneous O-Maxwell source equation, active whenever the conductor carries a current along an axial phase/lapse gradient $\partial_s \varphi$. The mechanism vanishes in the constant- φ , flat, symmetric Maxwell limit and is falsifiable through current-reversal, contact, and pulse-timing discriminants.

Observation. The three preconditions of that mechanism — an axial current channel, a non-zero axial $\partial_s \varphi$, and asymmetric boundary geometry — are jointly extremized in solar coronal magnetic flux tubes. Coronal loops carry field-aligned currents under the force-free / low- β condition; the chromosphere–transition region–corona stratification realises the steepest φ gradient in the solar atmosphere via the HQIV shell ladder $\varphi(m) = 2(m + 1)$; and the photospheric footpoints provide the asymmetric “contacts” (opposite-polarity, convectively-driven) that the wire paper identifies as the boundary-form revealing geometry.

Result. We map the wire-paper’s longitudinal stress to a coronal magnetic flux tube and obtain the boundary-form heating-flux readout

$$\frac{Q_{\parallel}}{A} = nq v_{\parallel} E_* \frac{3}{20\pi} \Lambda_s [\varphi(\text{corona}) - \varphi(\text{photosphere})],$$

with $\alpha = 3/5$ the lattice imprint, Λ_s the algebra-first axial coupling-log slot, and $E_* = 1$ in natural HQIV units. The full algebraic spine is formalised in the Lean module `Hqiv/Physics/CoronalLongitudinalStress.lean` [?], registering each equation as a `Prop` theorem with zero `sorry`.

The model is a *third* steady-state coronal-heating channel alongside Alfvén-wave dissipation and Parker-type nanoflare reconnection; it does not displace either. Its predictions are distinguished by being boundary-dominated rather than bulk, oriented under footpoint-polarity reversal, and dependent on the photosphere $\leftrightarrow$ corona shell-index gap rather than on bulk loop length.

1. SCOPE AND CLAIM DISCIPLINE

This note transfers the longitudinal-stress mechanism of [?] to the coronal-heating context. The wire-paper claim is restricted to a constrained, resistive many-body conductor; coronal loops are the natural large-scale realisation of that configuration in which the phase-gradient term should be *maximally* active rather than maximally suppressed.

We do not claim:

- that the standard Alfvén-wave [?] or Parker-type nanoflare [?] heating channels are wrong or unnecessary;
- that HQIV alone matches the observed heating-rate budget ($\sim 10^4 \text{ W/m}^2$ quiet Sun, up to $\sim 10^7 \text{ W/m}^2$ active region) without further evaluation of the φ profile from a horizon-shell model;
- that the shell indices ($m_{\text{ph}}, m_{\text{cor}}$) appropriate to the solar atmosphere are derived inside the present HQIV corpus — they are parameters of `CoronalColumnShells` in the Lean module and need an independent stellar-atmosphere shell-evaluation pipeline (analogue of `bulk.py+CLASS` for cosmology) to be closed quantitatively.

What we do claim is exactly what the Lean module proves: **the algebraic spine of a longitudinal HQIV/O-Maxwell heating channel on coronal flux tubes is well-defined, depends only on the existing $\alpha = 3/5$, $\gamma = 2/5$, and shell ladder $\varphi(m) = 2(m+1)$ inputs, and reduces to the standard Joule-only limit precisely when $\partial_s \varphi = 0$.**

2. WHY THE WIRE MECHANISM TRANSFERS TO CORONAL LOOPS

The wire paper requires three conditions for the longitudinal residual to be non-negligible. In coronal loops each is satisfied to an extreme degree:

- (1) **Axial current channel.** Coronal magnetic loops carry field-aligned currents under the low- β / near force-free condition $\nabla \times \mathbf{B} = \mu_0 \mathbf{J}$ with $\mathbf{J} \parallel \mathbf{B}$. The loop's magnetic flux tube plays the role of the wire.
- (2) **Non-zero $\partial_s \varphi$.** The chromosphere–transition region–corona stratification produces a density drop by a factor $\sim 10^6$ and a temperature rise by a factor $\sim 10^{2-3}$ over a column-thickness of $\sim 10^2$ km. In the HQIV shell ladder $\varphi(m) = 2(m+1)$, this stratification corresponds to a non-trivial shell-index gap and therefore a substantial $\Delta \varphi$ along the loop.
- (3) **Asymmetric “contacts.”** The photospheric footpoints of any active-region bipolar loop are anchored in regions of opposite magnetic polarity with independent convective driving (granulation, supergranular shearing). This is the geometric analogue of the wire paper's “asymmetric clamps” that reveal the integrated boundary term.

The wire paper notes that the residual is boundary-form-dominated when Λ_s is approximately constant along the conductor:

$$\int_0^L \Lambda_s(s) \partial_s \varphi(s) ds \approx \Lambda_s [\varphi(L) - \varphi(0)].$$

For a coronal loop with photospheric footpoint at $s = 0$ and coronal apex at $s = L$, this reads as a heating channel whose magnitude depends on the photosphere $\leftrightarrow$ corona φ jump, not on the loop length.

3. O-MAXWELL SOURCE EQUATION ON THE LOOP AXIS

We reuse the algebra-first inhomogeneous O-Maxwell equation in the form already wired to the Lean corpus through `Hqiv/Physics/OMaxwellAlgebraSeed.lean` and `Hqiv/Physics/ModifiedMaxwell.lean`. For axial component $\nu = s$ on the loop, stationarity gives

$$\sum_{\mu} F_{0\mu s} = 4\pi J_{0s} + \alpha \Lambda_s \partial_s \varphi, \quad \alpha = \frac{3}{5},$$

where Λ_s is the value of `algebraicMaxwellCouplingLog` for the axial slot. The longitudinal HQIV electric channel is therefore

$$E_{\text{HQIV}}(s) = E_* \frac{\alpha}{4\pi} \Lambda_s \partial_s \varphi(s) = E_* \frac{3}{20\pi} \Lambda_s \partial_s \varphi(s),$$

formalised as `coronalLongitudinalHQIVField` with the $\alpha = 3/5$ rewrite as `coronalLongitudinalHQIVField_alpha`

The total effective axial field $E_{\parallel}^{\text{eff}} = E_{\text{Ohm}} + E_{\text{HQIV}} = J/\sigma + E_{\text{HQIV}}$ is `coronalEffectiveAxialField`, and its classical-Maxwell collapse $E_{\text{HQIV}} \rightarrow 0$ when $\partial_s \varphi = 0$ is `coronalEffectiveAxialField_classical_limit`.

4. FORCE DENSITY AND HEATING-RATE DENSITY

The axial force density is $f_{\parallel} = nq E_{\parallel}^{\text{eff}}$ (`coronalLongitudinalForceDensity`), and the heating-rate density per unit volume is

$$\dot{q}(s) = f_{\parallel}(s) v_{\parallel}(s) = nq \left[\frac{J}{\sigma} + E_* \frac{3}{20\pi} \Lambda_s \partial_s \varphi(s) \right] v_{\parallel}(s),$$

formalised as `coronalHeatingRateDensity`. The Joule-only collapse with $\partial_s \varphi = 0$ is `coronalHeatingRateDensity` the HQIV-only specialisation with $J = 0$ is `coronalHeatingRateDensity_hqiv_only`.

Suppression of the Ohmic background in the coronal regime. Coronal Spitzer conductivity is enormous ($\sigma_{\text{cor}} \sim 10^6$ S/m or higher under typical conditions), so the field-aligned Ohmic axial field $E_{\text{Ohm}} = J/\sigma$ is correspondingly small. The HQIV channel does *not* go through $1/\sigma$; it is governed entirely by $\partial_s \varphi$ and the algebra-first Λ_s . The dimensionless ratio

$$\epsilon_{\text{HQIV}}^{\text{corona}} = \frac{E_{\text{HQIV}}}{E_{\text{Ohm}}} = \sigma A \cdot \frac{E_*(3/20\pi)\Lambda_s \partial_s \varphi}{I}$$

is therefore amplified relative to a laboratory copper wire by exactly the factor by which coronal field-aligned conductivity exceeds copper conductivity. This is the dimensionless statement of why the wire experiment is at the wrong end of the conductivity range to see the effect easily, while the corona is at the right end.

5. BOUNDARY FORM: PHOTOSPHERE $\leftrightarrow$ CORONA φ JUMP

Under the wire-paper's Λ_s -constant approximation, the integrated axial HQIV force across a column is the boundary form

$$\Delta F_{\parallel} = A n q E_* \frac{3}{20\pi} \Lambda_s [\varphi(\text{corona}) - \varphi(\text{photosphere})],$$

formalised as `coronalLongitudinalForceBoundary` with $\alpha = 3/5$ inlined, and equivalently as `coronalLongitudinalForceBoundary_eq_alpha_form` with the symbolic $\alpha/(4\pi)$ factor. The corresponding area-integrated heating flux is

$$\frac{Q_{\parallel}}{A} = n q v_{\parallel} E_* \frac{3}{20\pi} \Lambda_s [\varphi(\text{corona}) - \varphi(\text{photosphere})],$$

formalised as `coronalHeatingFluxBoundary`.

HQIV shell ladder closure. Using the existing HQIV closed form $\varphi(m) = 2(m+1)$ (`phi_of_shell_closed_form`), the photosphere $\leftrightarrow$ corona φ jump is

$$\Delta \varphi = \varphi(m_{\text{cor}}) - \varphi(m_{\text{ph}}) = 2(m_{\text{cor}} - m_{\text{ph}}),$$

proved in Lean as `coronalPhiJump_closed_form`. Sign: $\Delta \varphi \geq 0$ for $m_{\text{ph}} \leq m_{\text{cor}}$ (`coronalPhiJump_nonneg`); strictly positive for $m_{\text{ph}} < m_{\text{cor}}$ (`coronalPhiJump_pos_of_lt`); zero exactly when the shells coincide (`coronalPhiJump_zero_of_equal_shells`). The shell-indexed boundary force is then

$$\Delta F_{\parallel}^{\text{shells}} = A n q E_* \frac{3}{20\pi} \Lambda_s \cdot 2(m_{\text{cor}} - m_{\text{ph}}),$$

formalised as `coronalLongitudinalForceBoundaryShells_eq_two_shellGap`, and the area-integrated heating flux as `coronalHeatingFluxBoundaryShells_eq_two_shellGap`.

Open input. The specific shell indices ($m_{\text{ph}}, m_{\text{cor}}$) appropriate to the photosphere and corona are not derived inside the present HQIV corpus. They appear as the two natural-number fields of the Lean structure `CoronalColumnShells`. A separate stellar-atmosphere φ -profile evaluation (analogous in spirit to the `bulk.py`+CLASS pipeline for cosmology) is required to close this input to a numerical heating rate.

6. COMPANION FLUID MOMENTUM SOURCE ON THE LOOP AXIS

The wire paper carries the algebraic content as both an EM/source equation and a fluid/momentum-source expression. The latter is

$$g_{\text{vac},s} = -\frac{\gamma}{6} \left(\varphi \partial_s \delta \dot{\varphi}' + \delta \dot{\varphi}' \partial_s \varphi \right), \quad \gamma = \frac{2}{5}.$$

This is the 1D axial projection of the full 3-vector `hqivVacuumMomentumSource3` from `HQIVFluidClosureScaffold`. Formalised as `coronalAxialVacuumMomentumSource`, with $\gamma = 2/5$ inlined to give the prefactor

`-1/15 (coronalAxialVacuumMomentumSource_eq_minus_one_fifteenth)`. The identification with component 0 of `hqivVacuumMomentumSource3` (spatial gradients concentrated along the loop axis) is `coronalAxialVacuumMomentumSource_eq_hqivVacuumMomentumSource3.axis0`.

This gives the mechanical face of the heating: the EM channel of §3 and the fluid channel above feed the *same* $\varphi/\delta\theta'$ slot, so the energy injected by $f_{\parallel} \cdot v_{\parallel}$ and the momentum injected by $g_{\text{vac},s}$ are not independent.

7. DISCRIMINANTS

7.1. Current-reversal asymmetry (paper §7 item 1). The wire paper notes that thermal expansion and magnetic pinch scale as I^2 (even in current), while a boundary-dominated phase-gradient residual can carry an odd-in- I component when the conductor geometry is asymmetric. Formalised as the even/odd decomposition of arbitrary $F : \mathbb{R} \rightarrow \mathbb{R}$ under $I \mapsto -I$ (`evenInCurrent`, `oddInCurrent`). The wire-paper test polynomial

$$F(I) = a_2 I^2 + a_4 I^4 + b_1 I + b_3 I^3 + F_0$$

splits as

$$F_{\text{even}}(I) = a_2 I^2 + a_4 I^4 + F_0, \quad F_{\text{odd}}(I) = b_1 I + b_3 I^3,$$

proved as `coronalForceTestPoly_even_part` and `coronalForceTestPoly_odd_part`.

The coronal analogue is footpoint-polarity reversal of a chosen loop: in a bipolar active region one footpoint carries net upward current, the other net downward. Reversing the loop's net axial current direction (by tracking the loop through its polarity reversal during emergence/decay) should leave the thermal/pinch background even but flip the oriented HQIV-driven component. Quantitatively: any heating contribution scaling as $b_1 I + b_3 I^3$ is the HQIV signature; the rest is conventional. This is the cleanest experimental discriminant the model offers.

7.2. Boundary vs bulk (paper §7 item 2,3). Because the integrated HQIV channel is a boundary term in φ , the heating flux depends on $(m_{\text{cor}} - m_{\text{ph}})$ rather than on the bulk loop length L . Two loops with the same footpoint conditions but very different lengths should carry the *same* integrated HQIV heating flux per cross-sectional area. Bulk wave-dissipation predictions scale with L ; bulk Joule heating scales with L ; the HQIV boundary form does not.

7.3. Pulse timing (paper §7 item 4). The wire paper's timing discriminant transfers: the O-Maxwell phase channel responds on electrical / phase-equilibration time, not on Alfvén transit. Observable signature: heating signatures that precede the Alfvén crossing time of the loop. Distinguishes HQIV simultaneously from wave heating (Alfvén transit-bound) and from impulsive nanoflare reconnection (events with characteristic energy-release times $\sim$ minutes).

7.4. Material coherence / footpoint structure (paper §7 item 5). The fluid closure layer predicts sensitivity to coherence through the eddy-viscosity factor $C \in [0, 1]$ (`hqivEddyViscosity`). Coronal heating should correlate with magnetic-network coherence — supergranular boundaries, network vs internetwork, plage vs quiet Sun — beyond what is explained by ambient Joule and wave-dissipation models.

8. WHERE THIS SITS AMONG EXISTING MODELS

Channel	Steady?	Bulk-form?	Symmetry signature
Alfvén wave dissipation	yes (driven)	yes ($\propto L$)	even (oscillatory)
Parker-type nanoflare reconnection	no (impulsive)	yes (statistical)	even (event-counted)
HQIV longitudinal stress	yes	no (boundary)	odd in net loop current

The HQIV channel is steady-state and does not require either bulk dissipation tuning of incoming wave power or fine-tuned event-count statistics. It is the only one of the three that flips sign under footpoint-polarity reversal of a single loop.

9. LEAN WITNESS MAP

Every equation in this note is registered as a Prop theorem in `Hqiv/Physics/CoronalLongitudinalStress.lean` [?], with zero `sorry` and zero injected solar constants beyond the HQIV ladder. The mapping is:

Paper equation	Lean theorem / definition
$E_{\text{HQIV}} = E_*(\alpha/4\pi)\Lambda_s\partial_s\varphi$	<code>coronalLongitudinalHQIVField</code>
$E_{\text{HQIV}} = E_*(3/20\pi)\Lambda_s\partial_s\varphi$	<code>coronalLongitudinalHQIVField_alpha_3_5</code>
flat limit: $\partial_s\varphi = 0 \Rightarrow E_{\text{HQIV}} = 0$	<code>coronalLongitudinalHQIVField_zero_of_dphi_zero</code>
$E_{\parallel}^{\text{eff}} = J/\sigma + E_{\text{HQIV}}$	<code>coronalEffectiveAxialField</code>
$\dot{q} = nq E_{\parallel}^{\text{eff}} v_{\parallel}$	<code>coronalHeatingRateDensity</code>
Joule-only limit	<code>coronalHeatingRateDensity_classical_flat_limit</code>
$\Delta F_{\parallel} = Anq E_*(3/20\pi)\Lambda_s\Delta\varphi$	<code>coronalLongitudinalForceBoundary</code>
$Q_{\parallel}/A = nq v_{\parallel} E_*(3/20\pi)\Lambda_s\Delta\varphi$	<code>coronalHeatingFluxBoundary</code>
$\Delta\varphi = 2(m_{\text{cor}} - m_{\text{ph}})$	<code>coronalPhiJump_closed_form</code>
$\Delta\varphi \geq 0$ for $m_{\text{ph}} \leq m_{\text{cor}}$	<code>coronalPhiJump_nonneg</code>
$\Delta F_{\parallel}^{\text{shells}} \propto (m_{\text{cor}} - m_{\text{ph}})$	<code>coronalLongitudinalForceBoundaryShells_eq_two_shellGap</code>
$g_{\text{vac},s} = -\frac{\gamma}{6}(\varphi\partial_s\delta\dot{\theta}' + \delta\dot{\theta}'\partial_s\varphi)$	<code>coronalAxialVacuumMomentumSource</code>
$\gamma = 2/5 \Rightarrow \text{prefactor } -1/15$	<code>coronalAxialVacuumMomentumSource_eq_minus_one_fifteenth</code>
1D axial $\equiv$ component 0 of	<code>coronalAxialVacuumMomentumSource_eq_hqivVacuumMomentumSource3_axi</code>
<code>hqivVacuumMomentumSource3</code>	
even/odd-in- I decomposition	<code>evenInCurrent, oddInCurrent</code>
test polynomial parity split	<code>coronalForceTestPoly_even_part,</code> <code>coronalForceTestPoly_odd_part</code>
non-negativity of $Q_{\parallel}/A$	<code>coronalHeatingFluxBoundary_nonneg</code>
end-to-end heating witness	<code>CoronalHQIVHeatingWitness</code> (+ <code>qDot_eq_alpha_3_5_form</code>)

10. CONCLUSION AND FALSIFIABLE PREDICTIONS

The longitudinal HQIV/O-Maxwell stress mechanism, originally written down to explain a laboratory wire residual, transfers cleanly to coronal magnetic flux tubes and provides a candidate steady-state heating channel. The heating-flux readout

$$\frac{Q_{\parallel}}{A} = nq v_{\parallel} E_* \frac{3}{20\pi} \Lambda_s [\varphi(\text{corona}) - \varphi(\text{photosphere})]$$

is fully formalised in Lean and depends only on the existing HQIV inputs ($\alpha = 3/5$, $\gamma = 2/5$, $\varphi(m) = 2(m+1)$). It vanishes in the constant- φ , symmetric, flat Maxwell limit and so does not displace the standard Joule/wave/nanoflare descriptions; it adds a third, oriented, boundary-dominated channel.

Falsifiable predictions:

- (1) **Footpoint-polarity asymmetry.** A bipolar loop's HQIV heating contribution should reverse sign under net-current reversal; thermal and pinch backgrounds should not.
- (2) **Length-independence.** Two loops with the same footpoint shell anchoring but different bulk length should carry the same HQIV heating-flux per cross-sectional area, distinguishing the channel from bulk Joule and bulk wave dissipation.
- (3) **Sub-Alfvénic timing.** The HQIV channel's phase-response time precedes the loop's Alfvén crossing time. Fast-cadence heating events shorter than the Alfvén transit are not explainable as wave dissipation but are consistent with O-Maxwell phase equilibration.

- (4) **Network correlation.** Heating efficiency per loop should correlate with photospheric network coherence beyond what ambient wave or reconnection models predict, through the fluid-closure eddy factor $C \in [0, 1]$.
- (5) **Shell-anchor scaling.** Once a stellar-atmosphere φ -profile evaluation fixes $(m_{\text{ph}}, m_{\text{cor}})$, the predicted heating-flux is exactly $2(m_{\text{cor}} - m_{\text{ph}})$ times the per-unit- $\Delta\varphi$ coefficient — a single integer-gap scaling.

What remains is the input identified in §5: an HQIV stellar-atmosphere shell-evaluation pipeline that pins $(m_{\text{ph}}, m_{\text{cor}})$ for the solar photosphere and corona. With that input, the algebraic spine assembled in `Hqiv/Physics/CoronalLongitudinalStress.lean` reads off a numerical heating-rate prediction that can be compared directly against the observed 10^4 – 10^7 W/m² coronal heating budget.

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